

Junior Automation & Controls Technician

Three-Project Portfolio Roadmap - DIY Low-Voltage College Edition

PROGRAMME INFORMATION	
PROGRAMME WINDOW	7 September 2026 - 28 February 2027
PORTFOLIO	Exactly three substantial progressive projects
BUILD MODEL	One evolving low-voltage DIY training rig
TARGET ROLES	Junior automation, controls, PLC, instrumentation and E&I technician

PURPOSE

Prove competence through requirements, design, safe construction, logic, testing, troubleshooting, commissioning evidence and honest technical explanation.

Prepared for Emmanuel Januarie

College-style personal learning programme - not an accredited qualification

Role alignment and project progression

The portfolio is designed for Junior Automation Technician, Junior Controls Technician, Automation & Controls Technician, Instrumentation & Controls Technician, PLC Technician, E&I Technician and related entry-level roles.

Project	Technical progression	Build window
1 - Industrial Motor & Conveyor	Electrical fundamentals -> field inputs -> PLC logic -> motor/output control	5 Oct-15 Nov 2026
2 - Automated Process & Instrumentation	PLC -> instruments/signals -> sequencing -> process control -> actuators	16 Nov-27 Dec 2026
3 - Integrated Remote/Offshore Simulator	Integrated PLC -> HMI/SCADA -> networking -> faults -> commissioning	28 Dec 2026-28 Feb 2027

ONLY THREE PROJECTS

Do not split SCADA, networking, instrumentation, water treatment, troubleshooting or commissioning into extra standalone projects. They are integrated into Projects 2 and 3.

Single evolving rig

Project 1 establishes a reusable control station and conveyor representation. Project 2 reuses the control station and expands it with a DIY process/tank representation. Project 3 integrates and extends the same hardware and software into the flagship remote-process simulator.

Implementation boundary and honest claims

ACCEPTABLE EVIDENCE

- ☐ Battery/USB low-voltage circuits
- ☐ Breadboard and hobby components
- ☐ DIY frames, tanks and brackets
- ☐ PLC software simulation
- ☐ Microcontroller physical emulator
- ☐ Software-generated analog signals
- ☐ Laptop HMI/SCADA
- ☐ Safe open-circuit fault injection

DO NOT DIY

- ☐ Mains or three-phase wiring
- ☐ VFD power circuits
- ☐ Safety-rated E-stop/ESD systems
- ☐ Pressure-rated vessels
- ☐ Hazardous-area equipment
- ☐ Protection relied on for people
- ☐ Unenclosed mains power supplies
- ☐ Claims of industrial/offshore experience

REQUIRED WORDING

Each project must state: This low-voltage DIY and software simulation represents industrial control behavior for training. It is not certified industrial, safety, hazardous-area or offshore equipment, and it does not represent site authorization.

Evidence labels

Label	Meaning
REAL HARDWARE	A physical low-voltage device actually used
DIY-BUILT	A learner-made structure, panel, bracket, tank or trainer
HOBBY COMPONENT	A low-cost device representing an industrial function
SIMULATED	Behavior produced in PLC/process/network software
NOT TESTED	Claim or condition not performed
CONCEPT ONLY	Awareness without implementation

DIY-first evolving training rig

STARTING INVENTORY

Available now: breadboard, lamps/LEDs, resistors, wires and an alkaline battery. Confirm whether each lamp is an LED or bulb and its voltage/current before connection. Never connect an LED without a resistor or deliberately short the battery.

Buy only when the project requires it

Priority	Low-cost item	Purpose and reuse
1	One ESP32 or Arduino-compatible development board + USB cable	Optional physical I/O controller/emulator for all projects; do not call it an industrial PLC.
2	4 pushbuttons or DIP switches; 2 x 10 kohm potentiometers	Start/Stop/sensors/modes plus simulated level, pressure, flow or speed.
3	Logic-level MOSFET module + flyback diode	Interface a small DC motor or pump; never drive it directly from a controller pin.
4	Small 3-6 V DC hobby motor	Optional moving conveyor evidence.
5	Small low-voltage pump, tubing and temperature sensor	Optional Project 2 realism; LEDs/potentiometers remain acceptable substitutes.
6	Digital multimeter	Safe low-voltage measurement learning; do not DIY the measuring instrument.

Software path

Create industrial-style PLC logic in TIA Portal/PLCSIM if accessible, or another clearly identified IEC learning environment. A hobby controller may run the physical model separately. Project evidence must distinguish PLC simulation from microcontroller code. Use Ignition Maker or another personal learning HMI/SCADA platform when available.

Portfolio governance and release rules

EVERY PROJECT BEGINS WITH

- ☐ Problem statement and scope
- ☐ Real/DIY/hobby/simulated boundary
- ☐ Requirements and exclusions
- ☐ Architecture and interfaces
- ☐ Safety limits
- ☐ Acceptance criteria
- ☐ Versioned baseline

EVERY PROJECT ENDS WITH

- ☐ Normal/fault/recovery results
- ☐ Defect closure and regression
- ☐ As-built/source files
- ☐ Evidence index and README
- ☐ Technical demonstration
- ☐ Limitations and next steps
- ☐ Backed-up release

Common scoring rubric

Category	Weight	Release question
Requirements and design	15%	Is intended behavior explicit and traceable?
Technical implementation	25%	Does the safe model behave correctly and remain understandable?
Fault handling and safety	20%	Are abnormal states protected and recoverable within the training boundary?
Testing and evidence	20%	Do expected and actual results prove the claims?
Documentation	10%	Can another technician follow and maintain the work?
Demonstration	10%	Can decisions, faults and limitations be explained honestly?

RELEASE THRESHOLD

Target at least 75/100 with no category below 60%. Any uncorrected safety, ownership, stale-data or protective-response defect blocks release.

Project 1 - Industrial Motor & Conveyor Control System

BUILD WINDOW	PROJECT LABEL
5 October-15 November 2026	Low-voltage DIY / Simulation Training Project

PROJECT BRIEF

Build a small-scale, low-voltage industrial-style conveyor and motor-control system. The conveyor may be DIY-built from inexpensive materials; lamps/LEDs may represent actuators until a small motor is affordable.

Required competence

- [] Electrical fundamentals: voltage, current, resistance, Ohm's Law and power
- [] DC circuits, relays/basic control paths and circuit-protection concepts
- [] Motor control, digital inputs/outputs and sensor states
- [] PLC architecture, Ladder Logic, latching, timers and counters
- [] Auto/Manual, interlocks, permissives and controlled restart
- [] Fault handling, emergency-stop concept and PLC/electrical troubleshooting

Foundation chain

SIGNAL PATH

Electrical components -> wiring -> field inputs -> PLC -> Ladder Logic -> interface/output -> motor or process representation.

Project 1 - DIY implementation and requirements

Function	Low-cost implementation	Industrial meaning
Conveyor structure	Cardboard/wood/plastic frame, bottle-cap rollers or static diagram	Material transport mechanism
Motor	LED first; optional small DC hobby motor through MOSFET/driver	Motor/contactor/VFD-controlled load
Object sensors	Jumpers/buttons; optional light or proximity hobby sensor	Photoelectric/proximity field inputs
Start/Stop/Reset	Jumpers or low-cost pushbuttons	Operator control station
Modes	DIP switches or jumpers	Auto/Manual and Local/Remote ownership
E-stop concept	Normally closed simulated input only	Training representation, never safety-rated
PLC	TIA/PLCSIM or declared IEC software; optional hobby controller for physical I/O	Industrial controller behavior

Functional requirements

- ☐ Start/Stop with seal-in and safe stopped default
- ☐ Auto/Manual and Local/Remote ownership
- ☐ Entry/exit object detection and count target
- ☐ Run command, running feedback simulation and start-failure timer
- ☐ Jam/missing-object fault, overload representation and reset after correction
- ☐ Power-cycle/restart behavior that does not cause uncontrolled motion

Project 1 - documentation, tests and evidence

DOCUMENTATION

- ☐ Electrical/control schematic
- ☐ Wiring diagram
- ☐ PLC I/O and tag lists
- ☐ Control narrative
- ☐ Sequence description
- ☐ Test procedure/results
- ☐ Fault reports
- ☐ Photos/video and code screenshots

MINIMUM FAULTS

- ☐ Stop circuit open
- ☐ Missing start permissive
- ☐ Sensor stuck on/off
- ☐ Run command without feedback
- ☐ Overload/E-stop condition
- ☐ Timer/counter defect
- ☐ Incorrect I/O mapping
- ☐ Power-cycle recovery

Acceptance demonstrations

- ☐ Explain voltage/current path and calculate one real breadboard load
- ☐ Trace Start from physical/simulated input through PLC logic to output
- ☐ Run one complete manual and automatic cycle
- ☐ Show permissive and interlock preventing startup
- ☐ Inject a fault, diagnose by layer, correct it and verify recovery
- ☐ Present the difference between hobby hardware and the industrial target design

PROJECT 1 RELEASE

Release when electrical drawings, I/O, logic, actual behavior and fault evidence agree. The project establishes the reusable control foundation for Projects 2 and 3.

Project 2 - Automated Process & Instrumentation System

BUILD WINDOW	PROJECT LABEL
16 November-27 December 2026	Low-voltage DIY / Simulation Training Project

PROJECT BRIEF

Expand the Project 1 control foundation into a small low-voltage process: feed reservoir -> pump -> process tank -> controlled outlet/valve -> storage/output. DIY containers, tubing and supports are encouraged.

Required competence

- ☐ PLC automation with digital and analog I/O
- ☐ Level and temperature measurement; flow where practical
- ☐ Pressure concepts using simulation when necessary
- ☐ Sensors, transmitters, 4-20 mA concepts, scaling and calibration fundamentals
- ☐ Pump and valve/actuator control with sequencing
- ☐ PV/SP/feedback and PID fundamentals
- ☐ Manual/Automatic, alarms, interlocks, permissives and fault handling
- ☐ Instrument troubleshooting and loop-check evidence

MEASUREMENT HONESTY

Industrial transmitters are not required. Potentiometers, inexpensive sensors, signal simulators and software values may represent 4-20 mA and process variables. Clearly label each measurement REAL HARDWARE, HOBBY COMPONENT or SIMULATED.

Project 2 - DIY process and control implementation

Process element	DIY / low-cost option	Required behavior
Feed/process/storage tanks	Reused plastic bottles or containers in a leak tray	Visible process stages and safe containment
Pump	LED representation; optional small USB/low-voltage pump via driver	Command, feedback simulation, duty/fault response
Outlet valve	LED/servo representation or manual tubing clamp	Open/close command, position/failure concept
Level	Potentiometer plus independent jumper high/low switches	Continuous PV plus protective discrete limits
Temperature	Potentiometer first; optional low-cost sensor	Scaling, trend and alarm
Flow	Timed volume calculation, pulses from button or simulated value	Engineering-unit conversion and no-flow fault
Pressure	Software/potentiometer simulation	Range, scaling, abnormal-value and transmitter concepts

Control requirements

- [] Fill/transfer/drain sequence with explicit transitions
- [] Manual/Automatic and safe command ownership
- [] Low-low and high-high protection independent of routine control
- [] Pump command/feedback discrepancy and no-flow detection
- [] PV/SP display and basic closed-loop/PID simulation
- [] Alarm deadband, acknowledgement, reset and recovery rules

Project 2 - engineering documentation package

DESIGN DOCUMENTS	TEST EVIDENCE
☐ P&ID	☐ Calibration sheet
☐ Electrical/control schematic	☐ As-found/as-left values
☐ Instrument and I/O lists	☐ Loop/I/O check sheets
☐ Instrument tags	☐ Test procedure/results
☐ Wiring/loop diagrams	☐ Fault reports
☐ Control narrative	☐ Photos/videos
☐ Alarm list	☐ PLC/HMI evidence
☐ Cause/effect matrix	☐ Real/simulated register

Minimum instrument faults

- ☐ Disconnected or forced-low signal
- ☐ Abnormal analog value
- ☐ Incorrect scaling range
- ☐ Frozen/noisy value
- ☐ Pump fails to start
- ☐ Valve fails to move
- ☐ Missing permissive
- ☐ High-high trip and verified recovery

PROJECT 2 RELEASE

Demonstrate the chain: process -> instrument -> signal -> PLC -> control logic -> actuator -> process response. Reuse the control foundation and prepare the process rig for Project 3.

Project 3 - Integrated Remote/Offshore Process Control Simulator

BUILD WINDOW	PROJECT LABEL
28 December 2026-28 February 2027	Low-voltage DIY / Simulation Training Project

PROJECT BRIEF

Flagship project: integrate and expand Projects 1 and 2 into a low-voltage training simulator inspired by remote/offshore process equipment. It is not an oil rig and must not imply certified offshore, hazardous-area or industrial authorization.

Integrated process concept

PROCESS CHAIN

Feed -> pump -> process vessel -> instrumentation -> PLC/control -> separator/filter representation -> storage/output.

Flagship scope

- [] PLC digital/analog I/O, sensors and simulated transmitters
- [] 4-20 mA concepts, scaling and validation
- [] Pumps, motors, valves/actuators and process sequencing
- [] Manual/Automatic, interlocks, permissives, trips and ESD concepts
- [] Alarms, HMI, SCADA, trends and history
- [] PLC/HMI communications, Ethernet and network diagnostics
- [] Commissioning, FAT/SAT-style testing and structured fault-finding

Project 3 - integrated architecture

Layer	Reused/expanded implementation	Evidence
Physical process	Project 1 conveyor/motor representation plus Project 2 tank/process rig	Photos, labels, limitations and process diagram
Field devices	Buttons, jumpers, pots, hobby sensors and software values	Real/simulated register and loop checks
Control	PLC simulation and/or clearly separated hobby-controller I/O emulator	I/O list, logic, sequence and diagnostics
Supervision	Laptop HMI/SCADA with alarms, trends and historian concepts	Screens, tag quality, alarm and trend evidence
Network	Laptop loopback/Wi-Fi/Ethernet or software endpoints	Network diagram, IP/protocol map and fault evidence
Documentation	Coordinated engineering and commissioning pack	Revised final release

HMI/SCADA requirements

- [] Overview, equipment detail, alarm and trend navigation
- [] Command, mode, owner, permissive, feedback and fault visibility
- [] Bad/stale quality indication; no plausible frozen values presented as live
- [] Alarm priority, acknowledgement and operator response
- [] Useful trends for PV, SP, output, pump/valve status and faults

Project 3 - communications and fault campaign

Network requirements

- [] Labelled topology and address/protocol schedule
- [] PLC/HMI communications path and ownership
- [] Ethernet/IP fundamentals without falsely claiming proprietary EtherNet/IP implementation
- [] Modbus TCP/RTU or OPC UA concept/implementation where available
- [] Layered troubleshooting: physical, link, address, transport, protocol, mapping and application

Fault	Training implementation
Failed sensor	Stuck, disconnected or implausible field value
Abnormal analog value	Under/over-range or incorrect scaling
Failed pump	Command without feedback/process response
Valve failure	Command without position/process effect
Missing permissive	Start correctly blocked with visible cause
E-stop/ESD condition	Training input produces defined trip response
PLC logic fault	Controlled defect, change record and regression
HMI communication failure	Bad/stale quality and command inhibition
Network fault	Address/protocol/connectivity issue isolated by layer

REQUIRED TROUBLESHOOTING CHAIN

For every fault: Symptom -> initial checks -> evidence -> isolation -> diagnosis -> root cause -> correction -> verification -> documentation.

Project 3 - flagship documentation and demonstration

ENGINEERING PACKAGE	COMMISSIONING PACKAGE
☐ P&ID	☐ FAT/SAT-style procedure
☐ Electrical/control schematic	☐ Loop/I/O check sheets
☐ PLC I/O and instrument lists	☐ Commissioning punch list
☐ Loop diagrams	☐ Troubleshooting reports
☐ Control narrative	☐ Test and regression results
☐ Sequence description	☐ HMI/SCADA screenshots
☐ Cause/effect matrix	☐ PLC screenshots/code
☐ Trip/interlock matrix	☐ Final project report
☐ Alarm list	☐ 5-10 minute demonstration
☐ Network diagram	☐ Backups and evidence index

Demonstration sequence

Stage	What to show
1 - Scope	Low-voltage DIY/simulation boundary and process purpose
2 - Architecture	Field signal through PLC, HMI and actuator/process response
3 - Normal operation	Manual and automatic cycle with trends
4 - Protection	Permissive/interlock/trip preventing unsafe simulated behavior
5 - Fault	One deliberate fault diagnosed using evidence
6 - Recovery	Correction, verification and regression
7 - Close	Limitations, lessons, files and next improvement

FLAGSHIP RELEASE

The system must prove integration and disciplined troubleshooting, not imitate the appearance of a real offshore facility. Honest low-voltage evidence is stronger than exaggerated industrial claims.

Portfolio packaging and final release

FOLDER PATTERN

00_README | 01_Requirements | 02_Design | 03_PLC | 04_HMI_SCADA | 05_Network | 06_Testing | 07_Faults | 08_Demo | 09_Backups

Evidence index fields

- ☐ Evidence ID and project
- ☐ Document/file title
- ☐ Revision and date
- ☐ Real/DIY/hobby/simulated label
- ☐ Requirement or test linked
- ☐ What the artifact proves
- ☐ Known limitation
- ☐ File path and backup

Final release checklist

- ☐ Exactly three projects only
- ☐ One evolving low-voltage rig
- ☐ Every claim supported by evidence
- ☐ Drawings, tags, logic, HMI and tests agree
- ☐ Normal, fault and recovery demonstrated
- ☐ No unsafe or overstated industrial/offshore claims
- ☐ Flagship demonstration lasts 5-10 minutes

PORTFOLIO STATEMENT

These are personal low-voltage DIY and software training projects created to demonstrate junior automation and controls reasoning. They were not commissioned on a live industrial plant and do not represent certified safety, hazardous-area or offshore experience.